

JONEIL CAOILE

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PROFESSIONAL SUMMARY

M.S. Biomedical Engineering candidate (Stevens, 3.9 GPA) with a B.S. in Kinesiology and 400+ hours of clinical experience. Combines clinical intuition with engineering training to design and evaluate medical devices that work in real clinical workflow. Background in human-centered design, simulated-use study design, and clinician-facing dashboard development.

TECHNICAL SKILLS

Human Factors & UX: Human-centered design, usability testing, simulated-use workflow design, task analysis, clinical workflow analysis, IEC 62366 framework (coursework), data visualization, Figma for workflow mockups.

Research & Analysis: User interviews, IRB/HIPAA exposure (coursework), inter-rater reliability metrics, qualitative-to-quantitative synthesis, study protocol design, statistical analysis.

Technical: Python (NumPy, SciPy, Matplotlib), MATLAB, R, Qualtrics, SQL, Excel, Git; EMG signal analysis, motion capture (OptiTrack), biomechanics, cardiopulmonary physiology.

EDUCATION

Stevens Institute of Technology, *M.S. Biomedical Engineering* Expected May 2026 | GPA 3.9
Concentration: Medical Devices & Clinical Development | Provost Master's Scholarship

Relevant Coursework: Machine Learning, Clinical Research and Design, Biofeedback, Physiology, BMI, Biostatistics

California State University, Long Beach, *B.S. Kinesiology: Exercise Science* Dec 2020 | Cum Laude

PROJECTS

Chest Drain Air Leak Monitor – Inline Retrofit Cartridge (Advisor: Dr. Jinho Kim) Jan 2026 – May 2026 (Expected)

- Designed simulated-use workflows for an inline retrofit cartridge that adds digital air leak monitoring to existing chest drains, defining clinician interaction with the sensor outputs.
- Developed user-facing display concepts and alarm logic on the cartridge LCD, grounded in clinical scenarios and Cerfolio-sourced threshold logic.
- Communicated design tradeoffs and led scope cut decision based on clinician workflow priority.

Parkinson's Tremor-Suppression System (Product & Prototyping Lead) Sep 2025 – Dec 2025

- Designed a wearable EMG-based cuff concept for Parkinson's patients, prioritizing wearability, comfort, and usability for individuals with motor impairments.
- Applied usability principles to therapeutic wearable design; communicated tradeoffs to a multidisciplinary team grounded in measurable user outcomes.

Biofeedback Musculoskeletal Rehab Study (Study & Iteration Lead) Jan 2025 – May 2025

- Led a motion-capture study testing real-time audio feedback (Max/MSP) to improve knee alignment during Taekwondo kicks; functioned as usability testing for a biofeedback system.
- Operated OptiTrack motion capture, analyzed EMG and joint angle data from 3 athletes across 15 trials, and designed visualizations of activation patterns to support clinician decision-making.

EXPERIENCE

Graduate Researcher Jan 2025 – Present

Stevens Institute of Technology

- Design test protocols and instrumentation for a retrofit chest drain sensor module capstone in Dr. Jinho Kim's lab (Jan 2026–Present).
- Previously: Dr. Antonia Zaferiou's Musculoskeletal Control and Dynamics Lab (Jan-May 2025); see project work above.

Physical Therapy Aide 2016 – 2023 | 400+ Hours

Various Physical Therapy Clinics (Orthopedic, Pediatric, Geriatric)

- Assisted therapists with patient sessions, device setup, and treatment workflow across three clinical settings. Built first-hand understanding of end-user needs for medical device design and testing.

CERTIFICATIONS

Microsoft / LinkedIn Career Essentials (Apr 2026): Data Analysis, Cybersecurity (95%), Business Analysis (95%), Project Management; Atlassian Agile Project Management (94%); Aha! Product Management Certified.